AIR QUALITY DATA ANALYSIS (2004 - 2005)

Date

Select date range

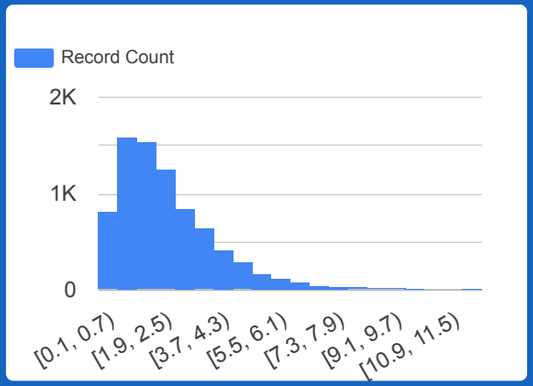
Record Count
9,357

CO(GT)
2.15

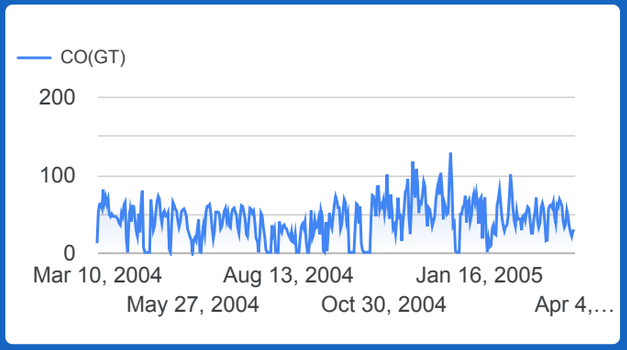
C6H6(GT)
10.08

NO2(GT)
113.08

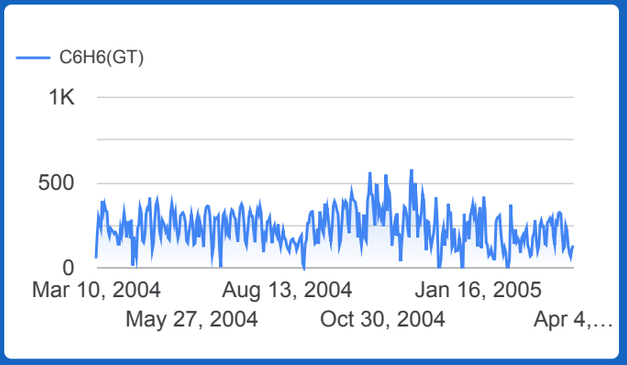
Distribution of CO(GT)



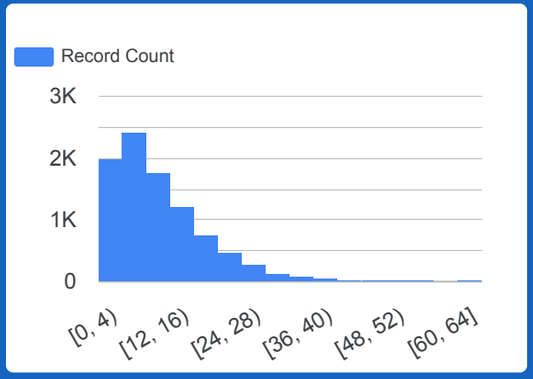
CO(GT) over time

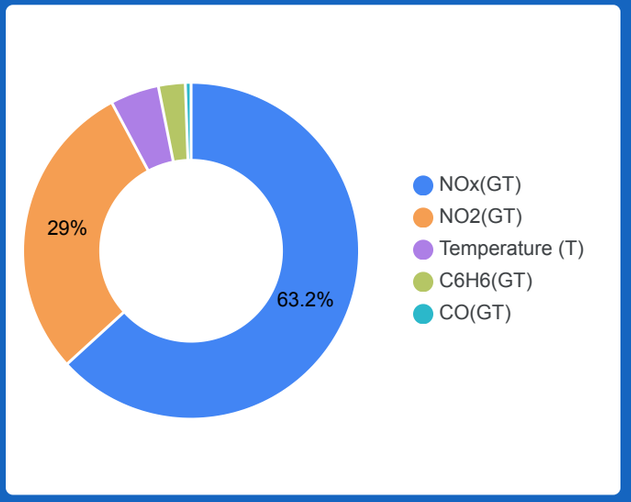


C6H6(GT) over time



Distribution of C6H6(GT)





KEY INSIGHTS

1. Average Pollutant Concentrations.The dataset recorded average concentrations of 2.15 for CO(GT), 10.08 for C6H6(GT), 246.88 for NOx(GT), and 113.08 for NO2(GT). These values provide an overview of the average pollutant levels recorded during the study period.

2. Hourly Variation in CO Concentration.
The exploratory analysis examined how average carbon monoxide (CO) concentration varied across different hours of the day. The hourly trend helps identify periods when CO concentrations were relatively higher or lower.

3. Pollutant Trends Over Time.
The line charts illustrate how selected pollutant concentrations, including CO, NOx and NO2, varied throughout the study period. These trends help identify changes in pollutant levels between different dates.

4. Distribution of Pollutant Concentrations.
Histograms were used to examine the distribution of CO(GT), PT08.S1(CO), and C6H6(GT). These visualizations help show how frequently different concentration ranges occurred and whether the data contained noticeable variations or unusual values.

5. Data Quality and Preparation.
Data cleaning was an important part of the analysis. Placeholder values represented by -200 were addressed, and duplicate records were removed to improve the reliability and consistency of the dataset used for the analysis.